

Introduction to Angular

why we use it, and the ideas behind it



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This is your introduction to Angular. We look at what Angular is, a few of the benefits it gives us, and why we should use it. Then we answer two questions that always come up: what is TypeScript, and what is ECMAScript. Finally, we install it and meet the Angular CLI.

1 Why should we use Angular?

Angular is one of the most popular JavaScript frameworks for creating web applications, made and maintained by **Google**. Many developers use it, so the obvious question is: why should **we** use it? The answer is in the improvements it gives us. Let us look at a few.

2 Performance

From a performance standpoint, Angular gives us **faster initial loads, efficient change detection, and improved rendering time**. In simple words, the app starts quickly and updates the screen quickly.

It is not just about speed. Angular also gives us better **modularity** (code split into clean, separate parts), **dependency injection** (a tidy way to share code and services), and **testability** (it is easy to test your app). If these technical terms do not fully make sense yet, do not worry, we will cover each one in detail in the coming classes.

Modern Angular has kept getting faster with every release. Today it is on version 22, and each version has focused on smaller, faster apps. Speed is not an accident with Angular, it is a design goal.

3 Mobile support

Angular is **designed with mobile in mind from the ground up**. Mobile realities like touch interfaces, small screens, and limited hardware are all considered in how Angular is built.

So with Angular we can build a **single application that works across both mobile and desktop**, without bolting on separate third-party tools just to make it work on phones.

4 Language choices

Angular gives us more than one language to write our app in. You can use plain **JavaScript**, or the newer versions of the JavaScript standard, and most of all you can use **TypeScript**.

Among all the choices, **TypeScript is the most popular**, and it is what we will use in this course. In fact, Angular itself is built using TypeScript. This raises two natural questions: what is TypeScript, and what is ECMAScript? Let us answer both.

5 What is ECMAScript?

The JavaScript language standard is officially called **ECMAScript**. Over the years, many versions have been released, each adding new features to the language.

- Older names you may still hear: **ECMAScript 6** was officially renamed **ES2015**, so those two terms mean the same thing.

- ES2015 introduced now-everyday features like **classes, modules, and arrow functions**.
- Since then, a new version has come out roughly every year, and today's browsers support these modern features directly.

In one line: ECMAScript is just the official name for the JavaScript language itself, and it gets a new version with new features every year.

6 What is TypeScript?

TypeScript is a free, open-source language developed by Microsoft. It is a **superset of JavaScript**, which means every bit of JavaScript is already valid TypeScript, and TypeScript adds more on top. A step called **compiling** turns your TypeScript into plain JavaScript that any browser can run.

The big thing TypeScript adds is **types**, plus object-oriented features you may know from Java or C++, like **classes, interfaces, and inheritance**. If you have done any object-oriented programming before, TypeScript will feel easy to pick up.

Because of all this, writing, maintaining, and refactoring Angular apps in TypeScript is a genuinely pleasant experience, which is why TypeScript has become the number one choice for Angular developers.

Do I need to learn TypeScript before this course? No. We will cover every TypeScript concept you need as we go along, step by step. Another bonus: TypeScript is fully supported in all the popular code editors, so you can pick whichever editor you like best.

7 Installing Angular

Three steps. Do each one and check it before moving on.

Step 1: Install Node.js

Angular runs on top of Node.js, so install it first. Go to nodejs.org and install the **LTS** (stable) version. Angular needs a recent Node, version 20.19 or newer.

CHECK IT WORKED

```
node -v
npm -v
```

Both should print a version number. Installing Node also gives you **npm**, the tool that downloads packages.

Step 2: Install the Angular CLI

Install the CLI once, globally, so you can use it from anywhere:

TERMINAL

```
npm install -g @angular/cli
```

CHECK IT WORKED

```
ng version
```

If it prints the version (today that is Angular 22), you are ready.

Step 3: Create and run your first app

TERMINAL

```
ng new my-first-app
cd my-first-app
ng serve
```

THEN OPEN YOUR BROWSER TO

```
http://localhost:4200
```

You will see your Angular app running. That is it, your first Angular application is live.

8 What is the Angular CLI?

The **CLI** (Command Line Interface) is Angular's official command-line tool. It is the thing you just used with `ng`. Instead of setting up projects and files by hand, the CLI does the heavy lifting for you: it creates the project, generates code, runs the app, and builds it for release.

In one line: the Angular CLI is a helper tool you run in the terminal that creates, builds, and runs your Angular project for you, so you focus on writing the app, not on setup.

The commands you will use most

- `ng new <name>` – create a brand new Angular project, fully set up.
- `ng serve` – run the app on your machine, with auto-reload when you save.
- `ng generate component <name>` (short: `ng g c <name>`) – create a new piece of the app for you, with all its files.
- `ng build` – package the app into final files, ready to put on a real server.
- `ng test` – run your tests.
- `ng version` – show which versions you have installed.

Why the CLI matters: it gives every Angular project the same clean structure and saves you from hours of manual setup. When you type one command and a whole working feature appears, correctly wired up, that is the CLI doing the boring work so you do not have to.

9 Remember this

Angular is a popular framework from Google for building web apps.

Performance: fast loads, fast updates, plus clean modularity, dependency injection, and easy testing.

Mobile: built for mobile and desktop from one single app.

Language: we write Angular in TypeScript, the most popular choice.

ECMAScript is the official name of JavaScript. **TypeScript** is JavaScript plus types, made by Microsoft.

Install: Node.js, then `npm install -g @angular/cli`, then `ng new` and `ng serve`.

The CLI is the `ng` tool that creates, runs, and builds your app for you.

10 Homework

Install it all yourself and get `ng serve` running. Open `localhost:4200` and see your app. If you hit an error, read it, search it, and try again. That struggle is the real learning.

In your own words, write one line each: what is ECMAScript, and what is TypeScript?

Try one CLI command: run `ng generate component hello` inside your project and see what files appear.

One question to think about: why do you think Angular gives us a CLI instead of making us set everything up by hand?